

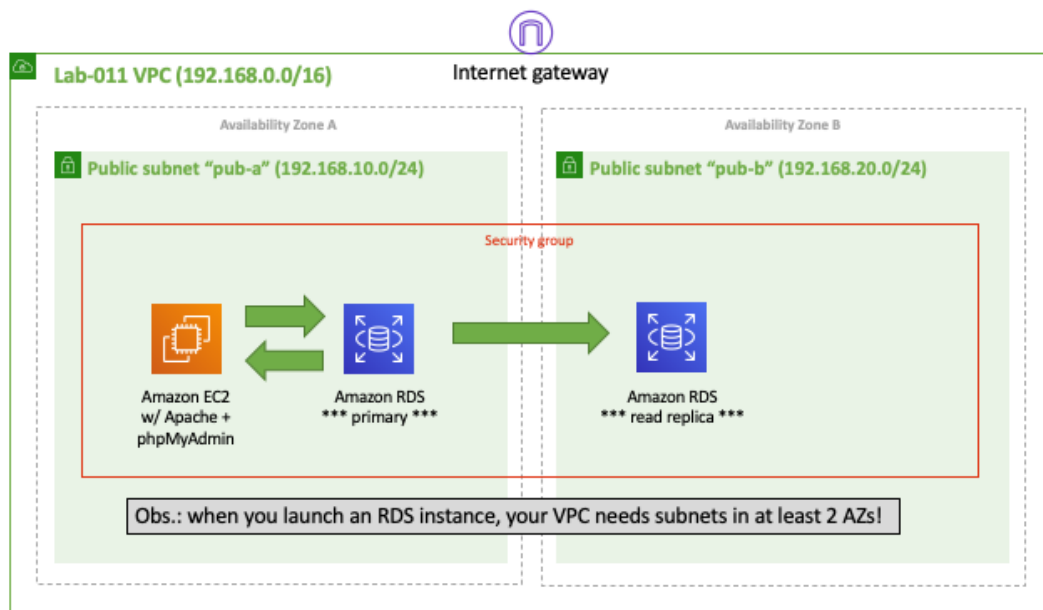
Lab-011

An RDS Instance with a Read Replica

Goal

This lab is similar to [lab-009](#) with the addition of a *read replica* RDS instance, a read-only copy of the database.

Architecture Diagram



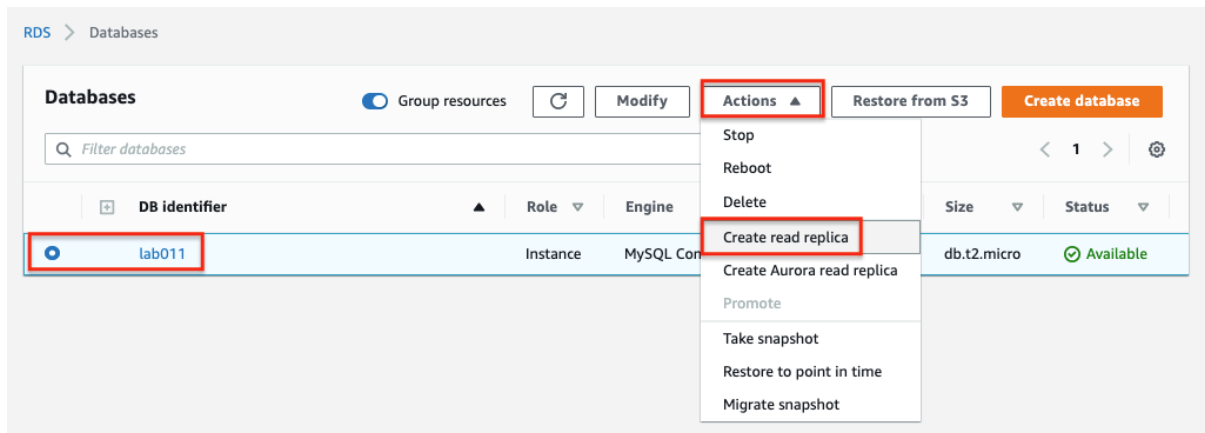
Overview

Follow the steps described in [lab-009](#) with the following changes when creating the RDS instance:

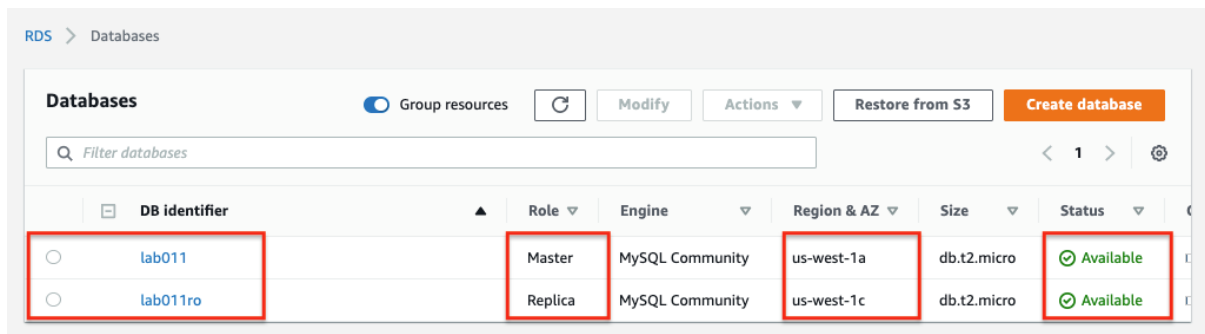
- Templates: Dev/Test (free-tier won't work this time)
- DB instance identifier: lab011
- DB instance size: select *Burstable classes* and pick the *db.t2.small*
- Connectivity: set your custom VPC to be the one where your RDS instance will run; in the additional connectivity configuration - VPC security group, choose *Create new* and name it *lab011sg*; finally for the availability zone choose the one suffixed with 1a (or the first option that appears to you)
- Additional configurations: set the initial database name to *lab011* and make sure *automatic backups* is enabled.

Step 1 - Create Read Replica

Select *Actions - Create read replica*.



Create the replica in a different AZ. Also, have it in the same security group *lab011sg*. For the *DB instance identifier* choose the name *lab011ro*. After a few minutes you should be able to see your *read replica* instance created.



Test and Validation

Have the phpMyAdmin configuration to first connect to the primary (master) database instance. Create a table (or modify an existing one). You should be able to do it.

Then try to connect to the read replica instead. Note that the read replica cluster has a different endpoint. You should be able to see the table modification that you did previously. Now try to *modify* the database (create a table or modify an existing one). You should not be able to do that now as the database is running in read-only mode.

Error

SQL query: [Copy](#)  [Edit](#)

```
INSERT INTO `Employees` (`id`, `name`) VALUES (
```

MySQL said: 

```
#1290 - The MySQL server is running with the --  
read-only option so it cannot execute this  
statement
```